

Patrick Pastorelli

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MSc in Computer Science from the University of Trento, specializing in low-power embedded systems and wireless communication. Co-author of peer-reviewed publications in embedded systems and robotics, including IEEE Sensors and IEEE RA-L. Experienced competitive programmer and SWERC 2022/2023 regional contestant.

Research Interests

My research interests lie at the intersection of embedded systems, wireless communications, and resource-constrained computing. I have research experience in designing intermittent computing architectures and energy-harvesting mechanisms for battery-free IoT devices, as well as developing efficient firmware for resource-constrained hardware. My work also includes low-power wireless communications, with hands on experience in wake-up radio protocols, UWB, and other wireless technologies. I am particularly interested in hardware-software co-design methodologies for constrained devices, focusing on developing efficient and reliable embedded systems that bridge low-level hardware capabilities with higher-level sensing and communication requirements.

Experience

IEEE RA-L

Invited Reviewer

2026–Present

Peer review of multiple Nonholonomic Motion and Path Planning manuscripts

University of Trento

Research Fellowship

2026–Present

Research on wake-up radios for energy-neutral systems, including implementation and characterization of a communication protocol on a new SoC.

University of Trento

Internship in PCB Design with Altium Designer

2025

Completed the PCB design cycle in Altium Designer, from schematic capture to multilayer layout and optimization.

University of Trento

Teaching Assistant in Computer Science

2025

Tutored undergraduate students in C++ debugging, memory management, and algorithmic problem solving; supported laboratory sessions and examinations.

University of Trento

Research Collaborator

Trento, Italy

2024–2025

Collaborated with Prof. Luigi Palopoli and Prof. Davide Brunelli on two main research tracks:

- Robotics: Developed a novel polyline smoothing algorithm for robot path planning, leading to a publication in IEEE RA-L.
- Embedded Systems: Main developer of the intermittent computing scheduler for the CapDYN smart battery system, coordinating computation under energy constraints, presented at IEEE Sensors 2025.

University of Trento and CovisionLab (Bozen, Italy)

Image compression pipeline in CUDA with CovisionLab

2024

Development and analysis of a high-speed image compression pipeline to accelerate data ingestion of AI models using CUDA. The project also included an evaluation of CUDA Optical Flow.

Publications

- 2026:** [Energy-Latency Characterization of an Integrated Wake-Up Radio SoC](#)
P. Pastorelli, D. Brunelli, and M. Nardello.
ENSsys 2026.
- 2025:** [Every Microjoule Counts: Zero-Failure Task Execution in Batteryless Sensors](#)
M. Nardello, M. Doglioni, S. Dagnino, **P. Pastorelli**, and D. Brunelli.
IEEE Sensors 2025.
Invited by IEEE Sensors Journal for a Special Issue on Selected Papers from the IEEE Sensors 2025 Conference
- 2025:** [Fast Shortest Path Polyline Smoothing With G1 Continuity and Bounded Curvature](#)
P. Pastorelli, S. Dagnino, E. Saccon, M. Frego and L. Palopoli.
IEEE Robotics and Automation Letters, Volume: 10, Issue: 4.
Also presented at IRIM 2024 and IROS 2025

Education

University of Trento

Trento, Italy

Master's Degree in Computer Science, 110/110 cum laude (GPA 4.0)

2023–2025

- Focus: Development and validation of firmware and drivers for batteryless intermittent computing and robotics, power profiling with oscilloscopes and logic analyzers, and simulation and hardware testing of wireless protocols and UWB based localization.
- Thesis: Exploring a Flooding-Based Wake-Up Radio Routing Protocol on a New Ultra-Low Power SoC for the IoT.

University of Genova

Genova, Italy

Bachelor's Degree in Computer Science, 110/110 cum laude (GPA 4.0)

2020–2023

- Focus: Team working, Operating Systems, Computer Architecture, Software Engineering.
- Thesis: Sorted containers: from Python to C++
- SWERC: Member of the first team at SWERC 2022/2023.

RUI Foundation

Genova, Italy

JUMP, Job-University Matching Project

2020–2023

Interdisciplinary course focusing on the development of transferable skills across three main areas: soft skills, modern themes, and interdisciplinary courses.

Technical Skills

Languages: Embedded C, C++23, Rust, Python, Bash, CUDA, Haskell

Embedded: ARM, RISC-V, STM32, ESP32, TI MSP430, FreeRTOS, ROS2, HAL/LL Drivers, PCB Design

Wireless: UWB, LoRa, IEEE 802.15.4, Sub-GHz, WURs

Tools & Systems: Linux, Git, Docker, LaTeX, MATLAB, Makefile

Others

Languages: Italian (Native), English (C1 level, FCE 182), Chinese (A1)

Interests: Former debater, also studied chess for 12 years, participating in multiple tournaments organized by FIDE such as the international chess festival of Imperia. Currently studying mandarin Chinese with the goal of obtaining an HSK certification in the future.